

Multi-Modal AI Strategy Builder

Onwuosiuno Ikhioda:

AI Product Manager - Customer Support Department

AIML-505-01R2K: Large Language/Generative AI

Dr. Lola Longe

Indiana Wesleyan University

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Introduction

Customer experience has become a major competitive advantage for organizations, making responsive and reliable customer support an important business objective. As the AI Product Manager, one of my department's primary goals is to improve customer satisfaction while reducing support response times and operational workload. Traditional customer support often results in delayed responses during off hours, long ticket queues, and inconsistent customer experiences. To address these challenges, I explored the use of multi-modal artificial intelligence to develop an AI-powered customer support chatbot capable of understanding text, screenshots, images, and voice inputs.

The proposed solution would provide customers with 24/7 assistance by answering common questions, guiding users through troubleshooting steps, assisting with onboarding and feature discovery, and escalating more complex issues to human support agents when necessary. To determine the most appropriate technology for this initiative, three AI platforms were evaluated: OpenAI's ChatGPT API, Google Vertex AI (Gemini), and Zendesk AI. Each tool was assessed using five evaluation criteria: effectiveness, ease of use, integration, cost, and ethical considerations.

Tool Evaluation

OpenAI (ChatGPT API)

OpenAI's ChatGPT API is a conversational AI platform that enables organizations to build intelligent applications capable of understanding and generating natural language. It also supports multimodal interactions, allowing customers to submit text alongside images such as screenshots to provide additional context when reporting issues. This makes it well suited for customer support environments where users need personalized assistance and detailed troubleshooting.

The platform's greatest strengths are its conversational accuracy, contextual understanding, and flexibility. It can be integrated into existing web and mobile applications through APIs and customized to align with an organization's knowledge base and support processes. However, implementing ChatGPT API requires software development resources, and usage costs increase as customer interactions grow. Additionally, organizations must implement safeguards to reduce inaccurate or misleading AI responses.

Overall, ChatGPT API received the highest ratings for effectiveness and integration due to its advanced conversational capabilities and extensive API support. It also scored well in ease of use because of its comprehensive documentation and developer resources. While the pricing model is flexible, organizations must carefully monitor API usage costs. From an ethical perspective, OpenAI provides enterprise security features and privacy controls, but organizations remain responsible for ensuring responsible AI deployment and protecting customer data.

Google Vertex AI (Gemini)

Google Vertex AI (Gemini) is Google's enterprise AI platform that supports multimodal processing across text, images, audio, video, and documents. Built on Google Cloud, it enables organizations to develop and deploy scalable AI applications while benefiting from Google's cloud infrastructure, security features, and machine learning services.

One of Vertex AI's primary advantages is its ability to analyze multiple forms of data within a single workflow, making it suitable for organizations handling complex customer interactions. It also provides strong security, governance, and compliance capabilities for enterprise environments. However, the platform is most beneficial for organizations already invested in Google Cloud, and its implementation can be more technically demanding than standalone AI APIs. Infrastructure and cloud usage costs may also increase as deployment scales.

Vertex AI scored highly in effectiveness, integration, and ethical considerations due to its robust multimodal capabilities and enterprise grade security. Ease of use was rated slightly lower because users must understand Google Cloud services to fully utilize the platform. Although pricing remains competitive, organizations should consider additional cloud infrastructure expenses when evaluating the total cost of ownership.

Zendesk AI

Zendesk AI is an AI-powered customer service platform designed to enhance existing customer support operations. Rather than serving as a standalone conversational model, it automates ticket routing, generates suggested responses, assists support agents, and integrates seamlessly with help desk workflows.

The platform's greatest strengths lie in its ease of implementation and operational efficiency. Organizations already using Zendesk can quickly adopt AI-powered automation with minimal

disruption to existing workflows. Zendesk AI also provides detailed analytics, reporting, and intelligent escalation of complex issues to human agents. However, it offers less flexibility for building highly customized AI experiences than general-purpose language models such as ChatGPT API, and many advanced AI features are only available through higher-tier subscription plans.

Zendesk AI received strong ratings for ease of use, integration, and ethical considerations because of its mature customer service ecosystem, role-based access controls, and enterprise security features. Although its effectiveness is high for automating customer support operations, its conversational capabilities are not as advanced as dedicated large language models. Its subscription-based pricing may also become costly for larger organizations requiring premium AI functionality.

Comparison and Recommendation

All three tools demonstrated strong capabilities for improving customer support, but each addresses different organizational needs. OpenAI's ChatGPT API offers the strongest conversational intelligence and flexibility, making it well suited for building a highly personalized AI assistant capable of understanding both text and image inputs. Google Vertex AI (Gemini) provides excellent multimodal capabilities and enterprise scalability, particularly for organizations that already operate within the Google Cloud ecosystem. Zendesk AI complements these technologies by providing customer service workflows, intelligent ticket routing, reporting, and seamless escalation to human agents.

Based on this evaluation, the recommended solution is a combination of OpenAI's ChatGPT API and Zendesk AI. ChatGPT API would serve as the conversational engine that delivers accurate, context-aware responses to customer inquiries, while Zendesk AI would manage support workflows, ticket creation, analytics, and agent collaboration. Together, these tools provide a scalable solution that improves response times, increases first-contact resolution, and enhances the overall customer experience while reducing the workload on support representatives.

Conclusion

This evaluation demonstrates that multi-modal AI could significantly improve customer support by providing faster, more personalized assistance through text and image understanding.

Implementing an AI-powered chatbot would reduce response times, improve customer satisfaction, and allow support agents to focus on resolving more complex issues. While all three platforms do provide valuable capabilities, combining OpenAI's ChatGPT API with Zendesk AI offers the strongest balance of conversational intelligence, operational efficiency, and scalability. This approach supports the department's objective of delivering a more responsive, efficient, and customer-centered support experience.

References

OpenAI. (2025). *OpenAI API documentation*. <https://platform.openai.com/docs>

Google Cloud. (2025). *Vertex AI documentation*. <https://cloud.google.com/vertex-ai/docs>

Zendesk. (2025). *Zendesk AI*. <https://www.zendesk.com/ai/>

Evaluation Criteria			
Evaluation Metric	OpenAI (ChatGPT API)	Google Vertex AI (Gemini)	Zendesk AI
Effectiveness	Excellent for conversational support, automated responses, onboarding, and knowledge retrieval. Highly customizable for product-specific use cases.	Excellent for multi-modal interactions, including analyzing screenshots, voice messages, and documents alongside text.	Excellent for customer service workflows, ticket routing, and agent productivity. Less flexible for broader product innovation.
Ease of Use	Moderate learning curve. Requires API integration and prompt design but offers extensive documentation.	Moderate to advanced. Best suited for organizations already using Google Cloud.	Easy. Designed for customer support teams with minimal technical expertise required.
Integration	Integrates with web apps, CRMs, help desk platforms, and custom applications through APIs.	Strong integration with Google Cloud ecosystem and enterprise applications.	Native integration with Zendesk and many CRM/customer service platforms.
Cost	Usage-based pricing. Costs scale with API usage and model selection. Additional expenses may include hosting, monitoring, and retrieval infrastructure.	Consumption-based pricing through Google Cloud. Additional costs for storage, AI services, and cloud resources.	Subscription pricing based on support plans and AI features. Premium capabilities may require higher-tier plans.
Ethical Considerations	Requires governance around customer data, prompt security, hallucination mitigation, and transparency regarding AI-generated responses.	Strong enterprise security features, but organizations remain responsible for privacy, compliance, and responsible AI practices.	Built with enterprise security features but still requires oversight regarding customer data handling, transparency, and escalation to human agents.

Overall Comparison

OpenAI (ChatGPT API)

Best For

- Personalized AI assistants
- Customer onboarding
- Knowledge assistants
- Conversational customer support

Advantages

- Highly flexible
- Excellent conversational quality
- Supports custom workflows

Course Companion	Google Vertex AI (Gemini) Best For • Multi-modal customer support • Image and document understanding • Enterprise-scale AI solutions Advantages • Excellent multi-modal capabilities • Strong enterprise integrations • Handles complex customer interactions involving images and voice Challenges • Greater implementation complexity • Higher infrastructure costs	Share
	Zendesk AI Best For • Organizations wanting rapid deployment • Existing customer service operations • Ticket automation Advantages • Easy implementation • Built specifically for support teams • Minimal customization required Challenges • Less flexibility outside customer service • Limited customization compared to general-purpose AI models Recommendation As the AI Product Manager, I would recommend a hybrid strategy: 1. OpenAI (ChatGPT API) as the core conversational AI engine to deliver personalized, context-aware support and onboarding experiences. 2. Zendesk AI to automate ticket routing, suggest responses, and streamline support operations if your organization already uses Zendesk. 3. Google Vertex AI (Gemini) for advanced multi-modal capabilities, such as analyzing screenshots, voice messages, and other rich customer inputs, particularly if your company operates within the Google Cloud ecosystem.	